ASSIGNMENT 04

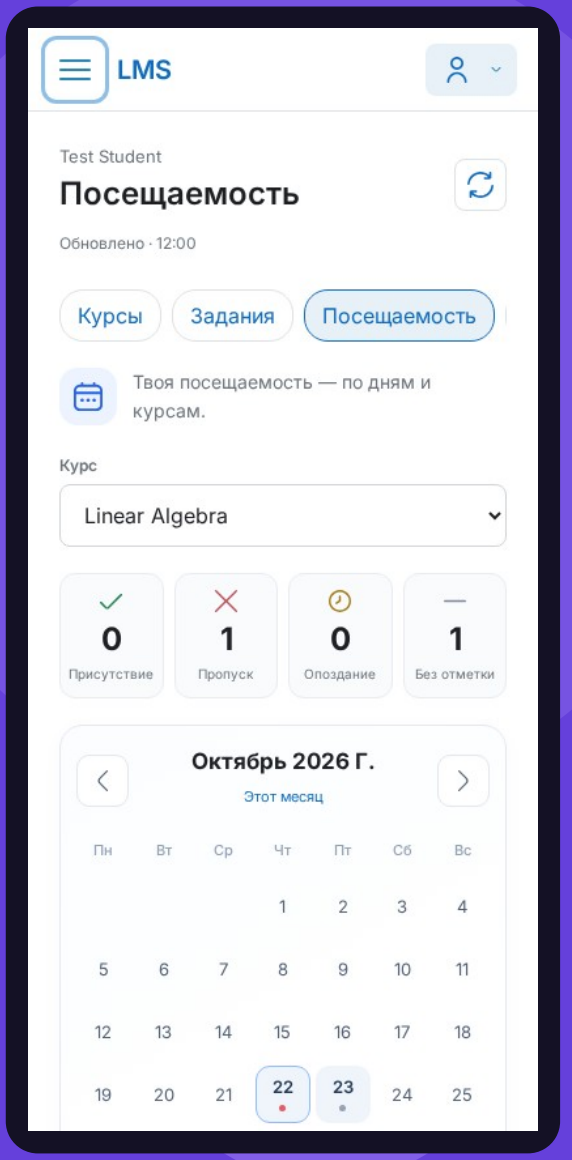
Multimedia Technologies

The media layer of My AITU

Formats, optimisation and accessible design for the LMS bridge prototype.

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Choose the format for the job

Three questions guide every choice: what is it, where is it used, and what must stay sharp?

MEDIA	DECISION TO MAKE	PROJECT APPLICATION
Images	Balance crisp interface details against file size.	WebP preview, PNG source and SVG icons
Audio	Separate distribution files from editing masters.	If narration is added: AAC / MP3; WAV for editing
Video	Choose both a container and a codec.	Planned: MP4 with H.264 video and AAC audio

01 QUALITY

02 FILE SIZE

03 COMPATIBILITY

One table. Ten media formats.

FORMAT	TYPE	COMPRESSION	ALPHA / MOTION	TYPICAL SIZE	SUPPORT	PROJECT USE
PNG	Image	Lossless	Alpha; APNG	Large photos	Broad	UI evidence
JPEG	Image	Lossy	None	Small photos	Broad	Photos
WebP	Image	Both modes	Alpha; motion	Compact	Modern	App preview
SVG	Image	Vector / text	Alpha; motion	Tiny icons	Broad	UI icons
MP3	Audio	Lossy	N/A	Small	Broad	Speech / music
AAC / M4A	Audio	Lossy	N/A	Small	Broad*	Narration
WAV / PCM	Audio	Uncompressed	N/A	Large	Broad*	Editing master
MP4 / H.264	Video	Usually lossy	Motion	Bitrate-based	Broad*	Demo delivery
WebM / VP9	Video	Usually lossy	Motion; alpha*	Bitrate-based	Modern*	Web playback
MOV / H.264	Video	Usually lossy	Motion	Bitrate-based	Limited web	Editing / transfer

Container = the package. Codec = how the media is encoded.

*Support depends on codec, OS and device. Size varies with content, resolution, bitrate and duration.
Sources cited in the original: MDN Image file type guide; Media container formats.

Same screen. Four encodings.

390 × 844 pixels in every file. Measurements use binary KiB: 1 KiB = 1024 bytes.

FORMAT / SETTING	SIZE (KiB)	VISUAL OBSERVATION
PNG	58.39	Sharp source
JPEG Q80	28.72	Minor edge artifacts
JPEG Q40	16.49	Visible edge artifacts
WebP Q80	15.75	Readable, compact

73.0%

smaller than PNG

WebP Q80 saves 42.64 KiB at the original resolution.

VIDEO EVIDENCE PENDING

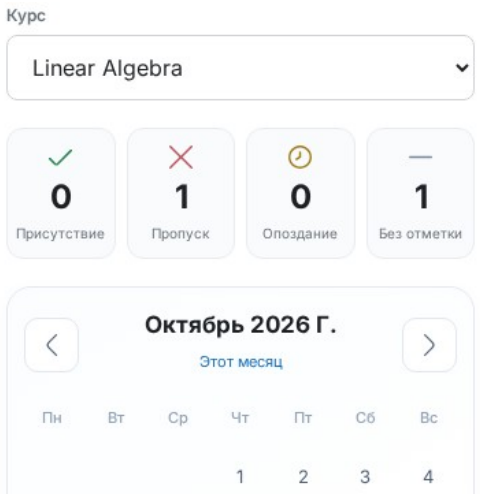
The same 10-second clip still needs two measured exports.

PNG / WebP: supplied files. JPEGs: exported from the same PNG at Q80 / Q40, 4:2:0 chroma, optimised, 25 Oct 2026.

Look at the edges, not just the number.

Identical 360 × 360 pixel crops, displayed at the same scale.

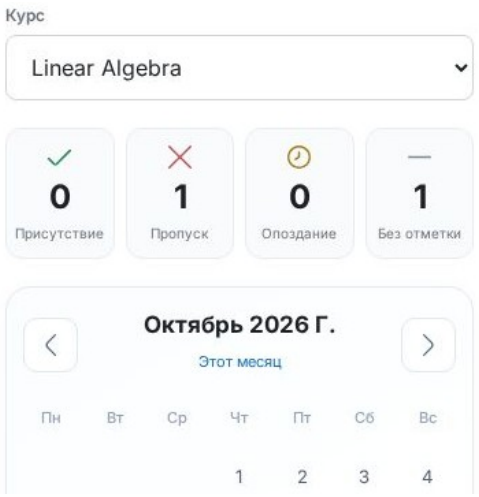
PNG



58.39 KiB

Crisp source text

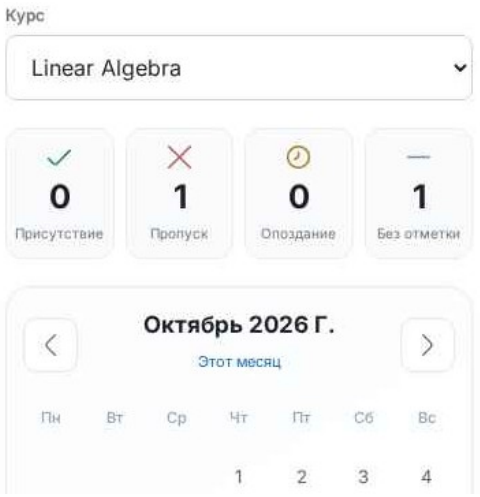
JPEG Q80



28.72 KiB

Light halos on edges

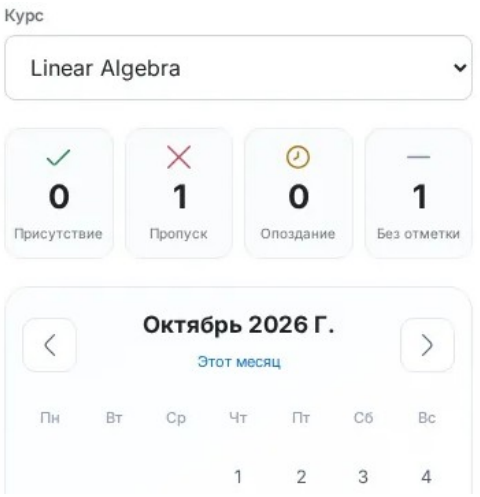
JPEG Q40



16.49 KiB

More obvious ringing

WebP Q80

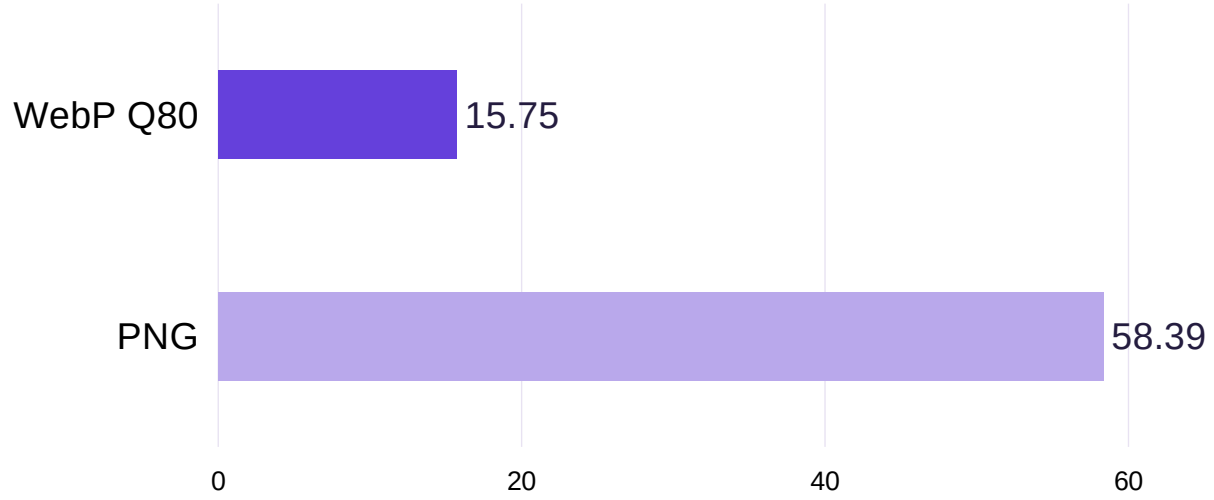


15.75 KiB

Readable UI details

Smaller files. Same screen dimensions.

Preview image size (KiB)



42.64 KiB saved per preview

1 **WebP Q80**

Use for the attendance preview.

2 **PNG**

Retain the lossless evidence image.

3 **Inline SVG + CSS**

Use vectors for icons and CSS for motion.

File-size reduction does not prove faster network loading.

18 icons. One HTML response.

Inline SVG markup is embedded in the page; the values below are source bytes, not separate downloads.

ASSET	WHERE USED	BYTES
3 · Book	Courses / nav	740
4 · Calendar	Tasks / attendance	1,162
5 · Graduation	Grades	857
6 · User	Account	869
7 · Menu	Navigation	561
8 · Close	Navigation	441
9 · Down	Account menu	391
10 · Sign-out	Account menu	906
11 · Refresh	Reload	945

ASSET	WHERE USED	BYTES
12 · Search	Course search	669
13 · Eye	Password visibility	935
14 · Check	Attendance	366
15 · Absent	Attendance	442
16 · Clock	Attendance time	777
17 · Minus	Unmarked status	284
18 · Left arrow	Previous month	416
19 · Next arrow	Next month	404
20 · List	Attendance status	1,010

Flaticon Uicons Regular Rounded · SVG viewBox 300 × 300 · Free licence with visible footer attribution.

The rest of the media stack

25 assets in the inventory: 18 icons and these seven visual, type and support elements.

ASSET	WHERE	FORMAT / DIMENSIONS	SIZE	SOURCE / LICENCE
1 · Attendance preview	Sign-in	WebP · 390 × 844	15.75 KiB	Own + icons[1]
2 · LMS wordmark	Sign-in	CSS / text · 68 × 68	Embedded	Own
21 · Login pulse	Login progress	CSS · 2 s loop	Embedded	Own
22 · Entrance motion	Transitions	CSS · 0.4 s	Embedded	Own
23 · Campus map	Map section	Responsive iframe	Variable	Yuujiso · MIT[2]
24 · Inter font	Interface	Web font · variable	Variable	Inter · OFL[3]
25 · Remote view	Optional support	JPEG Q70 · 1100 × 800	Variable	Private session

[1] Flaticon attribution in footer. [2] github.com/Yuujiso/aitumap. [3] github.com/rsms/inter.

The remote browser view is private and must stay out of the demo. Evidence PNGs and two unused SVG symbols are not loaded by the interface.

A lighter payload, with clear limits.

73.0%
smaller preview

30.8%
smaller controlled file set

0
expected image-request delta

CONTROLLED LOCAL PAYLOAD	PNG VERSION	WEBP VERSION	CHANGE
Preview image	59,787 B	16,126 B	−73.0%
4 front-end files + preview	141,601 B	97,941 B	−30.8%

Integrated

Responsive 250 px preview, inline SVG icons, CSS login pulse and reduced-motion support.

Still to measure

DevTools: transferred bytes, requests and Finish/Load time on Fast 4G with cache disabled.

Source: Part B inventory, 23 Oct 2026. Calculated local file bytes; external traffic and compression are excluded.

Make the experience understandable.

Implemented behaviours and checks from the prototype inventory.

IMPLEMENTED

Descriptive alt text

The attendance preview describes the calendar and status counts.

IMPLEMENTED

Decorative SVGs

Icons are hidden from screen readers; controls keep text or labels.

IMPLEMENTED

Reduced motion

The user’s preference disables animations and transitions.

IMPLEMENTED

Visible progress

Login progress has text and a visual pulse. No sound-only information.

NOT APPLICABLE

Captions / transcript

No speech audio or video is embedded in the prototype.

NOT APPLICABLE

Text over images

Captions sit below images; live text does not overlay raster images.

Show the product. Tell its story.

A narrated walkthrough of the My AITU / LMS bridge.

2:10

planned running time

01 **Sign in**

02 **Explore**

03 **Explain**

04 **Look ahead**

A clear role for every narrator.

Four scenes connect the working prototype to its purpose and next steps.

TIME	ON SCREEN	NARRATOR	TALKING POINT
0:00–0:30	Login screen	Nurassyl	Explain how sign-in works.
0:30–1:10	Main pages and navigation	Salima	Walk through content and design decisions.
1:10–1:40	Statistics and Stage 1 survey	Azhar	Explain the project’s purpose.
1:40–2:10	Demo sketches	Azhar	Outline the project’s future.

TOTAL · 2 MIN 10 SEC

Open the project demonstration.



My AITU / LMS bridge

Team walkthrough · Google Drive folder

OPEN VIDEO FOLDER

Scan to open

Opens the project's Google Drive folder.

The plan is defined. Export details remain.

Production note

The storyboard plans a 2-minute 10-second walkthrough of the My AITU / LMS bridge, with narration from all three team members. It covers sign-in, key pages, the project's purpose and future work. MP4 with H.264 video and AAC audio is the planned delivery format for broad device playback. The linked Drive folder is the hand-off location. Recording and editing tools, final resolution, frame rate, bitrate and file size were not supplied, so they cannot yet be reported as completed production settings. These values should be checked against the final exported video before submission.

FINAL FILE NOT INSPECTED

Confirm before submission

Recording + editing tools
Resolution + frame rate
Video + audio bitrate
Final duration + file size

From media assets to a working bridge.

The multimedia layer supports the same project and prototype.

01

Sign in

Individual user session

→

02

Read LMS

Course and attendance data

→

03

Present clearly

A responsive student interface

Run the project from its repository.

Linux launch sequence from bridge-test/README.md.

01

Clone the project

```
git clone https://github.com/d30mantcoder/ICTProject
```

02

Create an environment

```
cd ICTProject  &&  python -m venv venv
```

03

Install dependencies and Chromium

```
venv/bin/python -m pip install -r bridge-test/requirements.txt
venv/bin/python -m playwright install chromium --only-shell
```

04

Launch the bridge

```
BRIDGE_HEADLESS=1 venv/bin/python bridge-test/app.py
```

OPEN REPOSITORY

Local address
127.0.0.1:1740

Three changes that make the bridge useful.

CHANGE	WHY IT WAS NEEDED	CURRENT BEHAVIOUR
Real LMS data	A prototype needs useful, current course information.	Reads the user’s LMS content.
Interface redesign	Frequently used tasks were hard to reach.	Key actions are brought to the main page.
Asynchronous sessions	The earlier flow could not support independent users.	Users have separate sessions; slow requests do not hold others.

SCALE LIMIT STILL OPEN

<3 seconds per request at 1,000 users has not been verified.

One project. Three areas of ownership.

SZ

Salima Zimbekkyzy

PARTS C + D

Video editing

AS

Azhar Saparzholova

PARTS A + D

Multimedia format comparison

NO

Nurassyl Oralbekov

PARTS B + D

Media integration and optimisation

Less weight.
More clarity.

Multimedia choices that support a better student experience.

73%

smaller preview

18

inline SVG icons

25

inventoried assets